MGU-UGP (HONOURS) REGULAR EXAMINATIONS, MARCH 2026

Fourth Semester Discipline Specific Core Course (DSC)

MG4DSCIAM200 – PRINCIPLES OF ROBOTICS AND AUTOMATION

QP CODE: 25901339

(Answer Scheme)

Time: 1.5 Hours **Max. Marks: 50**

PART A MCQ (1 Mark Each)

(ANSWER ALL QUESTIONS)

1. **b) Amplify Control Signals** **[CO1]**
2. **b) Honda** **[CO1]**
3. **b) IR Proximity Sensor** **[CO1]**
4. **b) Windings** **[CO1]**
5. **b) Cannot Detect Object** **[CO1]**
6. **b) 3 Pins** **[CO2]**
7. **b) Relay** **[CO2]**
8. **b) Programmable Logic Controller** **[CO2]**
9. **c) Convert control signals into motion** **[CO3]**
10. **b) Timer** **[CO3]**

PART B (5 Marks Each)

(ANSWER ANY 4)

11. PIR sensor. Explain why it is called a passive sensor. **[CO1]**

- Definition – 2 marks
- Working Principle – 2 marks
- Reason for "Passive sensor" – 1 mark

12. Piezoelectric materials with examples. **[CO1]**

- Definition – 2 marks
- Examples - 1 mark
- Applications – 2marks

13. Servo motor and its basic purpose. **[CO1]**

- Definition – 2 marks
- Working - 2 marks
- Applications/ purpose – 1 mark

14. Types of I/O modules in PLC. **[CO2]**

- Definition of I/O – 1 mark
- Digital Input - 1 mark
- Digital Output – 1 mark
- Analog input/output – 2marks

15. Retentive timer & how it is reset. [CO2]

- Definition – 2 marks
- Working - 2 marks
- How timer gets Reset – 1mark

16. Ladder logic to stop the conveyor motor using an emergency push button [CO3]

- Ladder Diagram – 4 marks
- Explanation - 1 mark

PART C (10 Marks Each)

(ANSWER ANY 2)

17. Industrial Automation [CO1]

- Definition – 2 marks
- Productivity – 2 marks
- Safety – 2 marks
- Quality – 2 marks
- Examples - 1 mark

18. Sourcing and Sinking [CO2]

- Definition of sourcing – 3 marks
- Definition of sinking – 3 marks
- Diagram/ working - 1 mark
- Applications – 2marks

19. Pick and Place Robots [CO3]

- Definition – 2 marks
- Components – 3 marks
- Working process - 3 marks
- Diagram (just logic) – 2marks

20. Water Level Controller [CO3]

- Definition – 2 marks
- Working Principle - 4 marks
- Role of Controller - 2 mark
- Advantages / Applications – 2marks